



PASSAGE Solution Document

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Table of Contents

1. [Problem](#)
2. [Solution](#)
3. [System Overview](#)
4. [Transport Model](#)
5. [How the System Works](#)
 - Morning Ride Flow (Parent-Initiated)
 - Afternoon Ride Flow (System-Scheduled)
6. [PASSAGE Device \(Hardware\)](#)
 - Core Capabilities
 - Key Features
 - Interaction & Feedback Features
 - Device Behavior
7. [Identity & Verification](#)
 - Verification Rules (Mandatory)
8. [Backend Platform](#)
 - Core Responsibilities
 - Scheduling Logic
 - Administration & Management
9. [Rider App](#)
 - Core Functions
 - Trip Flow
10. [Parents App](#)
 - Core Features
 - Child Profile
 - Morning Ride Requests
 - Tracking
 - Notifications
 - Payments
 - Safety
11. [Safety & Emergency \(Cross-System\)](#)
12. [Power & Device Behavior](#)
13. [System Impact](#)
14. [Addendum A: PASSAGE Wearable – Hardware Components](#)
 - Alternative (Connectivity Option)
15. [Branding](#)
 - PASSAGE
 - Tagline
 - Vision
 - Mission
 - Core Values
 - What We Offer

- Brand Identity
 - Primary Colors
 - Accent Color Usage
 - Neutral Colors
 - Color Usage Principles
 - Typography
 - Logo Usage

1. Problem

In many urban areas in Uganda, particularly Kampala, parents of school-going children rely on informal transport such as boda bodas and taxis.

These systems are:

- Unstructured
- Lacking visibility
- Lacking accountability
- Missing standard safety controls

Existing alternatives such as boda bodas, taxis, school buses, or privately arranged rides are often expensive or not designed for children. Riders are not trained for child transport, and parents cannot track trips.

This creates gaps in safety, accountability, and reliability.

2. Solution

PASSAGE has created a structured school transport system that makes school transport safer, accountable, and visible.

It combines:

- A child wearable device
- A rider application
- A parent-facing platform

The system allows children to access rides without smartphones, while parents track and monitor each trip from pickup to arrival.

PASSAGE operates its **own network of trained and vetted riders**, assigned based on routes, schedules, and real-time requests.

3. System Overview

PASSAGE is a **technology-enabled child mobility system** that enables:

- Identity verification at pickup
- Real-time trip tracking
- Structured ride assignment

Core Components

- PASSAGE Device (child wearable)
- Backend Platform
- Rider App
- Parents App

4. Transport Model

PASSAGE operates a **hybrid ride model**:

Morning Rides (Home → School)

- Initiated by parents
- Requested via the parent app
- Rider assigned in real time or shortly after request

Afternoon Rides (School → Home)

- Automatically scheduled in the system based on school timetable
- Children assigned to riders based on routes
- No action required from parents

This model ensures:

- Flexibility in the morning
- Structure and efficiency in the afternoon

5. How the System Works

Morning Ride Flow (Parent-Initiated)

1. Parent requests ride via app (Pooled rides have to be requested at set times)
2. Backend assigns rider
3. Device wakes (triggered by request or manual interaction)
4. Rider arrives
5. Rider and child verify using:
 - Device ID
 - Rider identifier

6. Rider taps device to confirm pickup
7. Trip begins and parent is notified
8. Parent tracks trip in real time
9. At school, rider taps device to confirm drop-off
10. Parent receives confirmation
11. Device returns to sleep

Afternoon Ride Flow (System-Scheduled)

1. Backend pre-assigns rider and route
2. Device wakes automatically based on schedule
3. Rider arrives at school
4. Child is matched and verified
5. Rider taps device to confirm pickup
6. Trip begins and parent is notified
7. Parent tracks trip in real time
8. At home, rider taps device to confirm drop-off
9. Parent receives confirmation
10. Device returns to sleep

Note: No trip starts without successful verification.

6. PASSAGE Device (Hardware)

Core Capabilities

- GPS
- Motion sensor
- Cellular connectivity
- Encrypted data transmission
- NFC (tap-based communication for rider–child verification and trip start)

Key Features

- Lightweight, wearable, and durable
- Button-based interaction
- SIM / eSIM support
- Waterproof design

- Unique device ID for authentication
- Long-life rechargeable battery
- Auto sleep and auto wake
- Easy-to-use tactile controls
- Protective flip cover to prevent accidental input and protect the device screen

Interaction & Feedback Features

- Real-time proximity indicator to support rider–child coordination at pickup
- Provides continuous feedback (visual, vibration, or sound) as distance between rider and child reduces

Device Behavior

- Wakes when:
 - A ride is requested (morning)
 - A scheduled pickup time is reached (afternoon)
- Remains active during trip
- Returns to sleep after trip completion
- Sends location and verification data

[Refer to Addendum A: Bill of Materials for detailed hardware components.](#)

7. Identity & Verification

- Each child is linked to a unique device ID
- Riders do not see child names
- Verification uses **tap (NFC)**

Verification Rules (Mandatory)

Pickup:

- Rider taps device
- System confirms correct child and trip

Drop-off:

- Rider taps again
- Confirms safe arrival

No trip proceeds without these steps.

8. Backend Platform

The backend is the central system controlling operations.

Core Responsibilities

- Manage ride requests (morning)
 - Manage scheduled routes (afternoon)
 - Assign riders
 - Process verification
 - Handle tracking
 - Send notifications
 - Store trip data

Scheduling Logic

- Afternoon rides are pre-configured
- Riders are assigned based on clusters/routes for shared rides
- Device wake timing is synced with schedule

Administration & Management

- Manage school profiles and locations
- Register and manage children linked to schools and parents
- Onboard and manage riders (verification, assignment, status)
- Assign children to routes and riders
- Configure pickup and drop-off locations
- Monitor active trips and system activity
- Handle issue reporting and operational oversight

9. Rider App

Core Functions

- Receive assigned rides or routes
- View pickup sequence
- Perform tap-based verification

- Navigate to locations

Trip Flow

1. Assigned
2. En route
3. Arrived
4. Tap (pickup)
5. In transit
6. Tap (drop-off)
7. Completed

(All steps enforced)

10. Parents App

Core Features

Child Profile

- Add child details
- Set home and school locations
- Add guardians

Morning Ride Requests

- Request ride from home
- View assigned rider
- See estimated arrival

Tracking

- Live vehicle location
- Child status updates

Notifications

- Rider assigned
- Pickup confirmed
- Drop-off confirmed

- Delays or issues

Payments

- Prepaid wallet
- Trip cost visibility

Safety

- Emergency contact
- Trip history
- Guardian confirmation at drop-off

11. Safety & Emergency (Cross-System)

Device

- Emergency trigger (button or motion)

Backend

- Processes alerts
- Routes notifications

Apps

- Notify parents immediately

12. Power & Device Behavior

Device

- Executes sleep/wake
- Optimized for daily use

Backend

- Controls schedule timing (afternoon rides)

13. System Impact

PASSAGE:

- Makes school transport safer and more reliable
- Introduces accountability into informal systems
- Enables structured shared transport
- Improves affordability
- Gives parents visibility and control

Also creates a **data layer for urban mobility planning**.

14. Addendum A: PASSAGE Wearable – Hardware Components

Part Number	Purpose (Simple Description)
NANO-SIM-501	Holds the nano-SIM card and lets the device read it.
LIS2DH12TR	Measures motion and tilt along three axes.
PCF85063ATL/1,118	Keeps the date and time even when power is off.
TPS63031DSKR	Keeps power steady at 3.3V.
LQG15HN39NJ02D	39nH inductor for power or radio circuits.
LQW15AN9N1J00D	9.1nH inductor for power or radio circuits.
MLG1005S68NJT000	68nH inductor for filtering.
RE1C002UNTCL	N-MOSFET used to switch power.
RE1C001ZPTL	P-MOSFET used to switch power.
PMEG3010EJ,115	Protects circuits and manages small currents.
PRS2512-1R5NT	1.5uH inductor for power control.
RC0402FR-0715KL	15k Ω resistor.
RC0402FR-072KL	2k Ω resistor.
RC0402JR-071KL	1k Ω resistor.
SKRKAEE020	Small button for user input.
WM02N08F	N-MOSFET for switching power.
ABS07AIG-32.768kHz-7-T	Crystal that keeps timing accurate.
YC124-JR-07100KL	4-pack of 100k Ω resistors.
BSS138W-7-F	N-MOSFET for switching or level shifting.
4823-NRF9151-LACA-RCT-ND	Main cellular radio and MCU.
863-2032-1-ND	Boosts weak GPS signals.

Part Number	Purpose (Simple Description)
445-9081-1-ND	1uF capacitor.
TSM500P02CXRFGCT-ND	P-MOSFET for higher-current switching.
P17230CT-ND	10kΩ resistor.
256-W25Q64JVZPIQTRCT-ND	Flash memory for data and firmware.
535-14486-1-ND	GPS/RF antenna.
1.28 inch round LCD with touch	Display and touch input.
15-pin 0.5mm ZIF connector	Connects the display.

Alternative (Connectivity Option)

Part Number	Purpose (Simple Description)
497-ST4SI2M0020TPIFWCT-ND	eSIM chip (alternative to nano-SIM).

15. Branding

PASSAGE

Tagline

Safe mobility, built for children

Vision

To create a safe, structured, and trusted mobility system for children across cities where transport is informal and unregulated.

Mission

To make everyday school transport safer by building a system that guarantees verified pickup, accountable rides, and real-time visibility for every child.

Core Values

- Safety
- Accountability
- Reliability
- Accessibility

What We Offer

- Structured child transport
- Verified rider network
- Real-time tracking and visibility
- Secure pickup and trip verification
- Prepaid, controlled mobility system

Brand Identity

Primary Colors

- **Primary Color:** # 00bf63 (Green)
- **Secondary Color:** #5f6688 (Grey)
- **Accent Color:** #FFB300 (Amber)

- **Accent Color Usage**

The accent color is to draw attention to important states and interactions within the system. It is applied selectively to ensure clarity and avoid visual overload.

- **Primary Uses:**

- Alerts and warnings
- Verification prompts (e.g. child verification)
- Proximity and approach indicators (rider and child getting closer)
- Notifications and important status updates

Neutral Colors

- **White:** #FFFFFF
- **Light Grey:** #F5F7FA

Color Usage Principles

Green (#00BF63): Primary actions, confirmations, active states

Amber (#FFB300): Alerts, attention, proximity indicators

Grey (#5F6688): Structure, text, and icons

White (#FFFFFF): Background and spacing

Typography

- **Primary Font (Headings):** Inter (or Poppins as alternative)
- **Secondary Font (Body):** Roboto (or Open Sans as alternative)

Logo Usage

- **Primary Logo:** Green version (main usage)
- **Secondary Logo:** Grey version (subtle or document use)

Icons



Logos



PASSAGE



PASSAGE